MSG No. 140.544 $I4'/mc'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8,13,14,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[17,18,19,20,25,26,27,28]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,22,23,24,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,11,12,23,24,29,30]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[3,4,9,10,21,22,31,32]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,15,16,19,20,25,26]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,13,14,17,18,27,28]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6,9,10,13,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,4,7,8,11,12,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,21,22,25,26,29,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[19,20,23,24,27,28,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,5,6,19,20,23,24]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,7,8,17,18,21,22]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,10,13,14,27,28,31,32]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,15,16,25,26,29,30]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,3,21,23]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,4,22,24]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,7,17,19]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[6,8,18,20]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[9,12,29,32]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,11,30,31]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,16,25,28]`

8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[14,15,26,27]`

Table 6: Wyckoff site: `8f`, site symmetry: `4'..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,9,10]`
2	$\left[0, 0, \frac{1}{2} - z\right]$	`[3,4,11,12]`
3	$[0, 0, -z]$	`[5,6,13,14]`
4	$\left[0, 0, z + \frac{1}{2}\right]$	`[7,8,15,16]`
5	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[17,18,25,26]`
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[19,20,27,28]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[21,22,29,30]`
8	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[23,24,31,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2,23,24]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	`[3,4,21,22]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[5,6,19,20]`
4	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	`[7,8,17,18]`
5	$\left[\frac{1}{2}, 0, z\right]$	`[9,10,31,32]`
6	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	`[11,12,29,30]`
7	$\left[\frac{1}{2}, 0, -z\right]$	`[13,14,27,28]`
8	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	`[15,16,25,26]`

Table 8: Wyckoff site: `8h`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x, x + \frac{1}{2}, 0\right]$	`[1,6,19,24]`
2	$\left[-x, \frac{1}{2} - x, 0\right]$	`[2,5,20,23]`
3	$\left[x + \frac{1}{2}, x, \frac{1}{2}\right]$	`[3,8,17,22]`
4	$\left[\frac{1}{2} - x, -x, \frac{1}{2}\right]$	`[4,7,18,21]`
5	$\left[\frac{1}{2} - x, x, 0\right]$	`[9,14,28,31]`
6	$\left[x + \frac{1}{2}, -x, 0\right]$	`[10,13,27,32]`
7	$\left[x, \frac{1}{2} - x, \frac{1}{2}\right]$	`[11,16,26,29]`
8	$\left[-x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[12,15,25,30]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2`

No.	position	mapping
1	$\left[x, x, \frac{1}{4}\right]$	`[1,3]`
2	$\left[-x, -x, \frac{1}{4}\right]$	`[2,4]`
3	$\left[-x, -x, \frac{3}{4}\right]$	`[5,7]`

continued ...

Table 9

No.	position	mapping
4	$\left[x,\, x,\, \frac{3}{4}\right]$	`[6,8]`
5	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[9,12]`
6	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[10,11]`
7	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[13,16]`
8	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[14,15]`
9	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[17,19]`
10	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[18,20]`
11	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[21,23]`
12	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[22,24]`
13	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[25,28]`
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[26,27]`
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[29,32]`
16	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[30,31]`

Table 10: Wyckoff site: `16j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\, 0,\, \frac{1}{4}\right]$	`[2,12]`
3	$\left[0,\, x,\, \frac{1}{4}\right]$	`[3,9]`
4	$\left[0,\, -x,\, \frac{1}{4}\right]$	`[4,10]`
5	$\left[-x,\, 0,\, \frac{3}{4}\right]$	`[5,15]`
6	$\left[x,\, 0,\, \frac{3}{4}\right]$	`[6,16]`
7	$\left[0,\, -x,\, \frac{3}{4}\right]$	`[7,13]`
8	$\left[0,\, x,\, \frac{3}{4}\right]$	`[8,14]`
9	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{3}{4}\right]$	`[17,27]`
10	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{3}{4}\right]$	`[18,28]`
11	$\left[\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	`[19,25]`
12	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	`[20,26]`
13	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[21,31]`
14	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[22,32]`
15	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	`[23,29]`
16	$\left[\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{4}\right]$	`[24,30]`

Table 11: Wyckoff site: `16k`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,6]`
2	$[-x,\, -y,\, 0]$	`[2,5]`
3	$\left[y,\, x,\, \frac{1}{2}\right]$	`[3,8]`
4	$\left[-y,\, -x,\, \frac{1}{2}\right]$	`[4,7]`
5	$[-y,\, x,\, 0]$	`[9,14]`
6	$[y,\, -x,\, 0]$	`[10,13]`
7	$\left[x,\, -y,\, \frac{1}{2}\right]$	`[11,16]`

continued ...

Table 11

No.	position	mapping
8	$\left[-x,\ y,\ \frac{1}{2}\right]$	[12,15]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,22]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[18,21]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[19,24]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	[20,23]
13	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[25,30]
14	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[26,29]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[27,32]
16	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[28,31]

Table 12: Wyckoff site: 16l, site symmetry: ..m

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,24]
2	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[2,23]
3	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[3,22]
4	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[4,21]
5	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[5,20]
6	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[6,19]
7	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[7,18]
8	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[8,17]
9	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[9,31]
10	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[10,32]
11	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[11,29]
12	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12,30]
13	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[13,27]
14	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[14,28]
15	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[15,25]
16	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[16,26]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[3]
4	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[x,\ y,\ -z]$	[6]
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]
11	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[12]
13	$[y,\, -x,\, -z]$	[13]
14	$[-y,\, x,\, -z]$	[14]
15	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[19]
20	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[20]
21	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[23]
24	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[24]
25	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[26]
27	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[28]
29	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[31]
32	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[32]